Matgeo Presentation - Problem 5.9.8

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Problem Statement

A father's age is three times the sum of the ages of his two children. After 5 years his age will be two times the sum of their ages. Find the present age of the father.

Solution: Setting up Equations

Let the present age of the father be F. Let the sum of the present ages of his two children be S.

$$F = 3S \quad (1)$$

$$F - 3S = 0 \quad (2)$$

Solution: Equations after 5 years

After 5 years:

$$F + 5 = 2(S + 10) \tag{3}$$

$$F + 5 = 2S + 20 \tag{4}$$

$$F - 2S = 15 \tag{5}$$

Solution: Matrix Form

We now have a system of two linear equations. In matrix form $\mathbf{Ax} = \mathbf{b}$, this is:

$$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} F \\ S \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \end{pmatrix} \tag{6}$$

To solve this system, we use Gaussian elimination on the augmented matrix $[\mathbf{A}|\mathbf{b}]$.

$$\left(\begin{array}{cc|c} 1 & -3 & 0 \\ 1 & -2 & 15 \end{array}\right) \tag{7}$$

Solution: Gaussian Elimination

Perform the row operation $R_2 \to R_2 - R_1$:

$$\left(\begin{array}{cc|c} 1 & -3 & 0 \\ 0 & 1 & 15 \end{array}\right) \tag{8}$$

From the second row, we get:

$$S = 15 \tag{9}$$

Now, we perform back-substitution. Perform the row operation $R_1 \to R_1 + 3R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 45 \\ 0 & 1 & 15 \end{array}\right) \tag{10}$$

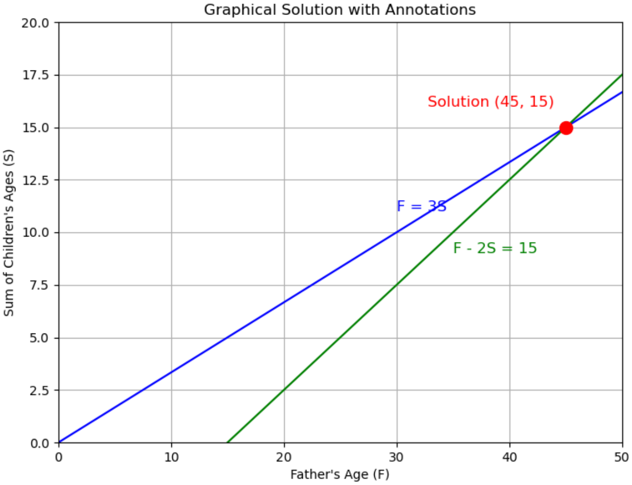
Solution: Final Answer

From the reduced row echelon form, we get the solution:

$$F = 45 \quad \text{and} \quad S = 15 \tag{11}$$

The question asks for the present age of the father. Thus, the present age of the father is **45 years**.

Graphical Solution



C Code: solver.c

```
#include <math.h>

int solve_system(double A[2][2], double b[2], double result[2]) {
    double det = A[0][0] * A[1][1] - A[0][1] * A[1][0];

    if (fabs(det) < 1e-9) {
        return 0;
    }

    double inv_det = 1.0 / det;
    double A_inv[2][2];
    A_inv[0][0] = A[1][1] * inv_det;
    A_inv[0][1] = -A[0][1] * inv_det;
    A_inv[1][0] = -A[1][0] * inv_det;
    A_inv[1][1] = A[0][0] * inv_det;

    result[0] = A_inv[0][0] * b[0] + A_inv[0][1] * b[1];
    result[1] = A_inv[1][0] * b[0] + A_inv[1][1] * b[1];

    return 1;
}
```

Python Code: main.py

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

lib = ctypes.CDLL("./libsolver.so")

lib.solve_system.argtypes = [
    (ctypes.c_double * 2) * 2,
    ctypes.c_double * 2,
    ctypes.c_double * 2
]
lib.solve_system.restype = ctypes.c_int

A = np.array([[1, -3], [1, -2]], dtype=np.double)
b = np.array([0, 15], dtype=np.double)

A_c = ((ctypes.c_double*2)*2)(*map(tuple, A))
b_c = (ctypes.c_double*2)(*b)
solution_c = (ctypes.c_double*2)()

status = lib.solve_system(A_c, b_c, solution_c)
```

Python Code: main.py (continued)

```
if status:
    fathers_age = solution_c[0]
    sum_childrens_ages = solution_c[1]

    print("Solution␣from␣C␣library:")
    print(f"Father's␣present␣age:␣{fathers_age:.0f}")
    print(f"Sum␣of␣children's␣present␣ages:␣{sum_childrens_ages:.0f}")

    F_vals = np.linspace(0, 50, 100)
    S1 = F_vals / 3.0
    S2 = (F_vals - 15.0) / 2.0

    plt.figure(figsize=(8, 6))
    plt.plot(F_vals, S1, color='blue')
    plt.plot(F_vals, S2, color='green')

    plt.plot(fathers_age, sum_childrens_ages, 'ro', markersize=10)

    plt.xlabel("Father's␣Age␣(F)")
    plt.ylabel("Sum␣of␣Children's␣Ages␣(S)")
    plt.title("Graphical␣Solution")
    plt.grid(True)
    plt.xlim(0, 50)
    plt.ylim(0, 20)
    plt.show()

else:
    print("Could␣not␣solve␣the␣system.")
```

Python Code: verify.py

```
import numpy as np

A = np.array([[1, -3], [1, -2]])
b = np.array([0, 15])

try:
    solution = np.linalg.solve(A, b)
    fathers_age = solution[0]
    sum_of_childrens_ages = solution[1]

    print("---␣Verification␣using␣NumPy␣---")
    print(f"Father's␣present␣age:␣{fathers_age:.0f}")
    print(f"Sum␣of␣children's␣present␣ages:␣{sum_of_childrens_ages:.0f}")

    assert abs(fathers_age - 45) < 1e-9
    assert abs(sum_of_childrens_ages - 15) < 1e-9
    print("\nResult␣is␣correct.")

except np.linalg.LinAlgError:
    print("The␣matrix␣is␣singular.␣NumPy␣could␣not␣find␣a␣unique␣solution.")
```